classification_report#

https://scikit-learn.org/stable/modules/generated/sklearn.metrics.classification_report.htm

Description:

Build a text report showing the main classification metrics. Read more in the User Guide. Ground truth (correct) target values. Estimated targets as returned by a classifier.

Function Signature

```
sklearn.metrics.classification_report(y_true, y_pred, *,
labels=None, target_names=None, sample_weight=None,
digits=2, output_dict=False, zero_division='warn')[source]#
```

Parameters:

y_true1d array-like, or label indicator array / sparse matrix

y_pred1d array-like, or label indicator array / sparse matrix

labelsarray-like of shape (n_labels,), default=None


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digits=2, output_dict=False, zero_division='warn')[source]#
```

Parameters:

y_true1d array-like, or label indicator array / sparse matrix

y_pred1d array-like, or label indicator array / sparse matrix

labelsarray-like of shape (n_labels,), default=None

Parameters

`y_true1d array`

Ground truth (correct) target values.

`y_pred1d array`

Estimated targets as returned by a classifier.

`labelsarray`

Optional list of label indices to include in the report.

`target_namesarray`

Optional display names matching the labels (same order).

`sample_weightarray`

Sample weights.

Code Examples

```
{'label 1': {'precision':0.5,
             'recall':1.0,
             'f1-score':0.67,
             'support':1},
 'label 2': { ... },
  ...
}
```

```
>>> from sklearn.metrics import classification_report
>>> y_true = [0, 1, 2, 2, 2]
>>> y_pred = [0, 0, 2, 2, 1]
>>> target_names = ['class 0', 'class 1', 'class 2']
>>> print(classification_report(y_true, y_pred,
target_names=target_names))
              precision    recall  f1-score   support

     class 0       0.50      1.00      0.67         1
     class 1       0.00      0.00      0.00         1
     class 2       1.00      0.67      0.80         3

    accuracy                           0.60         5
   macro avg       0.50      0.56      0.49         5
weighted avg       0.70      0.60      0.61         5

>>> y_pred = [1, 1, 0]
>>> y_true = [1, 1, 1]
>>> print(classification_report(y_true, y_pred, labels=[1, 2,
3]))
              precision    recall  f1-score   support

           1       1.00      0.67      0.80         3
           2       0.00      0.00      0.00         0
           3       0.00      0.00      0.00         0

   micro avg       1.00      0.67      0.80         3
   macro avg       0.33      0.22      0.27         3
weighted avg       1.00      0.67      0.80         3
```

Notes & Warnings

> **NOTE** See also precision_recall_fscore_supportCompute precision, recall, F-measure and support for each class. confusion_matrixCompute confusion matrix to evaluate the accuracy of a classification. multilabel_confusion_matrixCompute a confusion matrix for each class or sample.

Detailed Documentation

Documentation

Build a text report showing the main classification metrics.

Read more in the User Guide.

Ground truth (correct) target values.

Estimated targets as returned by a classifier.

Optional list of label indices to include in the report.

Optional display names matching the labels (same order).

Number of digits for formatting output floating point values. When output_dict is True, this will be ignored and the returned values will not be rounded.

If True, return output as dict.

Added in version 0.20.

Sets the value to return when there is a zero division. If set to “warn”, this acts as 0, but warnings are also raised.

Added in version 1.3: np.nan option was added.

Text summary of the precision, recall, F1 score for each class. Dictionary returned if output_dict is True. Dictionary has the following structure:

The reported averages include macro average (averaging the unweighted mean per label), weighted average (averaging the support-weighted mean per label), and sample average (only for multilabel classification). Micro average (averaging the total true positives, false negatives and false positives) is only shown for multi-label or multi-class with a subset of classes, because it corresponds to accuracy otherwise and would be the same for all metrics. See also precision_recall_fscore_support for more details on averages.

Note that in binary classification, recall of the positive class is also known as “sensitivity”; recall of the negative class is “specificity”.

Compute precision, recall, F-measure and support for each class.

Compute confusion matrix to evaluate the accuracy of a classification.

Compute a confusion matrix for each class or sample.

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Label Propagation digits: Active learning

class_likelihood_ratios